

MOBILE STRATEGY DOCUMENT

# Mobile PWA Roadmap

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A complete strategy for transforming BarberBoost into an installable mobile experience for barbers and customers on both Android and iOS — without building a new codebase.

# Approach Options

Three viable paths were evaluated for delivering BarberBoost on Android and iOS. Each was assessed against the criteria of development speed, code reuse, and native experience quality.



## Option 1 — Progressive Web App (PWA)

**RECOMMENDED****LOWEST COMPLEXITY**

Enhance the existing Next.js application to be installable on home screens like a native app. No new codebase is required — everything already built carries over. A `manifest.json`, service worker, and web push notifications are added on top of the current app.

**Trade-off:** No App Store or Google Play listing. Android (Chrome) has excellent PWA support including push notifications and install prompts. iOS 16.4+ supports push notifications via PWA; older iOS versions do not. Some users won't know how to install without a store link — mitigated by an in-app install prompt on both platforms.



## Option 2 — Capacitor

**MIDDLE GROUND**

Wraps the existing Next.js web app in a native shell and publishes to both the Apple App Store and Google Play Store. The codebase remains unchanged — Capacitor provides a bridge for native APIs including push notifications on both platforms.

**Trade-off:** UI is rendered in a WebView so it won't feel fully native on either platform, but it achieves store presence on both. Approximately 1–2 days of configuration. A natural upgrade path from the PWA if store listings become a business requirement.



## Option 3 — Expo (React Native)

**V2 RECOMMENDATION**

A fully native application using the existing API routes as its backend. All UI components are rewritten in React Native. SDKs for Supabase and Stripe are available. Approximately 70% of business logic (hooks, types, API calls) can be shared.

**Trade-off:** Requires building an entirely new frontend. Recommended as V2 once V1 PWA has validated mobile demand and revenue impact.

# V1 Feature Priorities

The V1 release targets both barbers and customers simultaneously, with a hard scope boundary to avoid shipping two half-finished products. Build order: Barber features first, then Customer features.



## Barber / Shop Owner

Build first — drives subscription retention

- **Today's Schedule**  
Clean list of the day's bookings. The #1 reason a barber opens the app.
- **Booking Status Updates**  
Mark as completed, no-show, or cancelled without a laptop.
- **New Booking Push Notification**  
Instant alert when a customer books. Critical for shops not at a desk.
- **Quick Client Lookup**  
Search by name, view visit history and contact info.
- **Today's Stats Glance**  
Bookings today and revenue today. Single screen.
- **Inventory Management** V2  
Deferred to V2
- **AI Marketing Copy** V2  
Deferred to V2 (Empire plan only)



## Customer

Build second — drives bookings to barbers

- **Book an Appointment**  
Service → Staff → Time → Confirmation. The entire value proposition.
- **Shop Direct Link**  
Existing /booking/[slug] flow. Most customers arrive via a shared link.
- **My Bookings**  
View upcoming and past appointments via lightweight guest lookup.
- **Appointment Reminder Push**  
"Your appointment is tomorrow at 2pm." Justifies installing the app.
- **Shop Search / Map Browse** V2  
Requires significant new backend work
- **In-App Payments** V2  
Adds App Store review complexity
- **Loyalty / Rewards** V2  
Not in current system

# Mobile Responsiveness Audit

Every key page and component was reviewed against small-screen behaviour. The codebase has stronger mobile foundations than typical Next.js dashboard apps — the shell, navigation, and most page layouts are already ready.

| Component                   | Status      | Finding                                                                                                                                                              |
|-----------------------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dashboard Shell             | • Ready     | BottomNav with safe-area insets, mobile sidebar drawer with backdrop, pb-20 content padding to clear nav.                                                            |
| Bottom Navigation           | • Ready     | 5 primary destinations, 44px minimum hit targets, env(safe-area-inset-bottom) for notched devices. Excellent.                                                        |
| Header                      | • Ready     | Hamburger menu on mobile, page title truncates, "New Booking" collapses to icon-only on small screens.                                                               |
| Dashboard Page              | • Ready     | All stat grids use grid-cols-1 sm:grid-cols-2 lg:grid-cols-4. Chart row stacks on mobile via grid-cols-1 lg:grid-cols-5.                                             |
| Bookings Page               | • Ready     | Week view hidden on mobile (hidden sm:contents). List and Day views are fully responsive. Controls collapse to icon buttons.                                         |
| Clients Page                | • Ready     | Responsive grid, horizontally scrollable tag filters, full-width search on mobile, list header hidden on small screens.                                              |
| Public Booking Page         | • Ready     | Explicitly mobile-first. Sticky "Book Now" bar on mobile, max-w-2xl centred layout, scrollable horizontal carousels. Best-in-class.                                  |
| Weekly Calendar (Dashboard) | • Needs Fix | Uses min-w-[600px] — technically scrollable but a 7-column time grid on a 390px phone is unusable. Should be hidden below lg, showing TodaySchedule only.            |
| Analytics Page              | • Needs Fix | BusiestHoursHeatmap and StaffTable have fixed-width columns that overflow on small screens. Wrap in overflow-x-auto containers or provide mobile-collapsed variants. |
| PWA Infrastructure          | • Not Built | No manifest.json, no service worker, no install prompt, no push notification support. This is the primary gap between "mobile website" and "installable PWA".        |



# PWA Implementation Plan

Four sequential phases take BarberBoost from responsive website to installable PWA with push notifications. Every improvement benefits the web app simultaneously.

1

## Installable — Manifest & Icons

~Half Day

- Create `/public/manifest.json` with `#c9a84c`, `#0a0a0a`, `standalone` —  
`name`, `background`, `display`, `consumed`  
`short` by both  
`name`, `Android`  
`theme` and iOS  
`colour`
- Generate icon assets at required sizes: 192×192, 512×512, and a maskable variant (Android adaptive icons); plus Apple touch icon variants for iOS home screen
- Add Android meta tags to `app/layout.tsx`: `theme-color`, `manifest link`, `mobile-web-app-capable`
- Add iOS meta `app/layout.tsx`: `apple-mobile-web-`, `apple-`, `apple-mobile-web-app-`  
`tags to` `app-capable` `touch-icon` `status-bar-style`
- Add an in-app install `beforeinstallprompt` event (Android/Chrome) with a manual "Add to Home Screen" guide for iOS Safari prompt using the

2

## Offline Fallback — Service Worker

~Half Day

- Install `next-pwa` (wraps Workbox, works with Next.js App Router)
- Configure in `next.config.ts` — precaches static assets automatically
- Add a simple `/offline` page rendered when the network is unavailable
- Verify service worker registration and caching in Chrome DevTools

3

## Mobile Layout Fixes

~1 Day

- Hide the `BookingCalendar` below `lg` — `TodaySchedule` only on mobile (single class change)  
`weekly` `show`
- Wrap analytics heatmap and staff table in `overflow-x-auto` scroll containers
- Audit inventory and settings pages for any additional overflow issues
- Verify all pages across iPhone SE (375px) and standard iPhone (390px) viewports

4

## Push Notifications

~2 Days

- Generate VAPID key pair, store public/private keys as environment variables
- Create `push_subscriptions` table in Supabase (device endpoint, keys, user/shop association)
- Add `/api/push/subscribe` API route to save device subscriptions

- Add subscription prompt component in DashboardShell — triggered on first mobile visit after install
- Trigger push from Stripe webhook on new booking, and from booking API on customer confirmation
- Customer reminder: trigger from existing cron reminder logic

### IMPLEMENTATION SUMMARY

**3.5**

Days total estimated effort

**0**

New codebases required

**100%**

Existing API routes reused

# Next Steps

The following actions are ready to begin immediately. All implementation has been planned against the existing codebase and requires no architectural changes.

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1

## Phase 1 — Create the PWA manifest and iOS meta tags

Add `/public/manifest.json`, generate icons for both Android (192×192, 512×512 maskable) and iOS (Apple touch icon variants), and update `src/app/layout.tsx` with Android `theme-color` and iOS `apple-mobile-web-app-capable` meta tags. This makes the app installable on both platforms immediately.

2

## Phase 2 — Add service worker with next-pwa

Install `next-pwa`, configure Workbox caching strategy in `next.config.ts`, and add an offline fallback page. Verify precaching and offline behaviour in Chrome DevTools.

3

## Phase 3 — Fix the two mobile layout issues

Hide `BookingCalendar` on mobile screens (one-line fix), and wrap analytics charts in `overflow-x-auto` containers. Verify layout across 375px and 390px viewports.

4

## Phase 4 — Implement push notifications

Set up VAPID keys, create the `push_subscriptions` database table, build the subscribe API route, add the in-app permission prompt, and wire push triggers into the booking and webhook flows.

5

## V2 Decision — Evaluate Capacitor for App Store presence

After the PWA has been live for 4–6 weeks, review mobile usage metrics and subscriber feedback. If App Store listing becomes a business requirement, migrate to Capacitor — it wraps the same codebase with approximately 1–2 days of additional configuration.